

Problems

Status

Standings

Little Sub and Triangles

Tags:

Time Limit: 2 s Memory Limit: 256 MB
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Codes

AI Code Tips

Description

Little Sub loves triangles. Now he has a problem for you.

Given n points on the two-dimensional plane, you have to answer many queries. Each query require you to calculate the number of triangles which are formed by three points in the given points set and their area S should satisfy $l \leq S \leq r$.

Specially, to simplify the calculation, even if the formed triangle degenerate to a line or a point which $S = 0$, we still consider it as a legal triangle.

Input

The first line contains two integer n, q ($1 \leq n \leq 250, 1 \leq q \leq 100000$), indicating the total number of points.

All points will be described in the following n lines by giving two integers x, y ($-10^7 \leq x, y \leq 10^7$) as their coordinates.

All queries will be described in the following q lines by giving two integers l, r ($0 \leq l \leq r \leq 10^{18}$).

Output

Output the answer in one line for each query.

Samples

input	Copy
<pre>4 2 0 1 100 100 0 0 1 0 0 50 0 2</pre>	
output	Copy
<pre>3 1</pre>	

Author

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Codes

HZNUOJ is based on HUSTOJ

--- Maintainer List ---

Server Time: 2025-8-26 16:08:11